SQL CASE & Derived Columns – Simplified Questions

1. 1. Create a new column that combines customer_first_name and customer_last_name as full_name from the customer table.
2. 2. Show a derived column named total_cost by multiplying quantity and cost_to_customer_per_qty in the customer_purchases table.
3. 3. List all customers along with the LENGTH() of their first name.
4. 4. Display all vendor names and vendor_type, and return a new column vendor_type_length.
5. 5. Use TRIM() to clean up product_size values and return only those that are not blank.
6. 6. Return the top 10 products where the trimmed size is not empty.
7. 7. Create a column is_zip_valid that returns 1 if customer_zip > 10000 else 0.
8. 8. Combine booth_number and booth_type into one string column named booth_info.
9. 9. Add a column price_per_item to vendor_inventory as a derived column from original_price / quantity.
10. 10. Show a full_name column in vendor combining owner first and last names.
11. 11. List purchases and create a new column high_quantity that returns ‘Yes’ if quantity > 5 else ‘No’.
12. 12. Add a season_flag column to market_date_info that shows 1 if market_season is ‘Summer’ or ‘Winter’, else 0.
13. 13. Return vendors with a new column that shows Fresh Produce if vendor_type contains ‘fresh’, else Other.
14. 14. Show customers where first name starts with 'A' and add a column starts_with_a with value 'Yes' or 'No'.
15. 15. Add a derived column to customer purchases that rounds total price to 2 decimals.
16. 16. Create a name_length_score by summing the lengths of first and last names of each customer.
17. 17. Show product_name and return a new column with uppercase product_qty_type.
18. 18. Return vendor inventory with a flag column if original_price is above 100.
19. 19. Add a column in market_date_info that says Cold if market_max_temp < 15, Moderate if between 15-30, else Hot.
20. 20. Display customer full names and flag them as Long Name if length > 15, else Short Name.
21. 21. Use TRIM() + LENGTH() to find products with empty or blank product_size.
22. 22. Add a column zip_group in customer where zip < 20000 = 'A', < 40000 = 'B', else 'C'.
23. 23. Use CASE to bin original_price in vendor_inventory into: 'Low', 'Medium', 'High'.
24. 24. Add a flag weekend_flag to market_day — 1 if it's Saturday/Sunday, else 0.
25. 25. Create a derived column vendor_rank — 1 if vendor_id < 5, 2 if between 5–10, else 3.
26. 26. Combine market_day and market_season as one column: day_season_label.
27. 27. Use CASE in product to say Perishable if category_id in (1, 2), else Non-Perishable.
28. 28. Add a column purchase_type in customer_purchases: Bulk if quantity > 10, Regular if 5–10, Light if < 5.

29. 29. For each customer, create a name_validity column: Valid if name has length > 3, Invalid otherwise.
30. 30. Return market dates with a column: event_flag = 1 if special_notes is not null.
31. 31. Add a rain_or_snow column in market_date_info using CASE.
32. 32. Create a column price_level_numeric: A = 1, B = 2, C = 3 in booth table.
33. 33. Show vendors where trimmed vendor_name has more than 10 characters.
34. 34. Return customers whose customer_last_name ends with 'ez' and length < 6.
35. 35. Return a transaction_band column in customer_purchases: Under ₹50, ₹50–₹100, Over ₹100.
36. 36. Create a customer_type column: Local if ZIP < 30000, Metro if ZIP between 30000–70000, Premium if above.
37. 37. Show a derived field in vendor_inventory indicating cost per item, rounded to 1 decimal.
38. 38. Combine booth_number, vendor_id, and market_date into a single assignment_code.
39. 39. Add a column to vendor called is_family_business = 1 if owner last name = vendor last name.
40. 40. Add category_flag to product: 1 if category_id is in top 3 by ID.
41. 41. Return purchase_summary that says: "Expensive Bulk" if quantity > 5 and price > 100, "Affordable" otherwise.
42. 42. Use CASE to define market_category: Rainy Day if rain flag = 1, Snowy Day if snow flag = 1, Clear Day otherwise.
43. 43. Build a seasonal_description column by combining season + day (e.g., "Summer Saturday").